

Course Code : ECT 457-1

QTSR/RW – 24 / 1809

**Eighth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

REAL TIME OPERATING SYSTEMS AND KERNELS

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All the questions are compulsory and carry marks as mentioned.
 - (2) Draw neat and clean diagram, wherever necessary.
 - (3) Use of Non – programmable calculator is permitted.
 - (4) Assume suitable data wherever necessary.
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1. (a) The services and functions provided by an operating system can be divided into two main categories. Briefly describe the two categories, and discuss how they differ. 5(CO1)
(b) Elaborate different types of Operating Systems. 5(CO1)
 2. (a) The first known correct software solution to the critical-section problem for two processes was developed by Dekker. The two processes, P0 and P1, share common variables : With reference to these Processes State and explain Dekkers algorithm. 5(CO2)
(b) Elaborate semaphore in OS with up and down mode. 5(CO2)
 3. (a) Differentiate in between pipes and message queues. 5(CO3)
(b) Explain Real-Time Clocks and System Clocks. 5(CO3)
 4. (a) A certain computer provides its users with a virtual memory space of 2^{32} bytes. The computer has 2^{22} bytes of physical memory. The virtual memory is implemented by paging, and the page size is 4,096 bytes. A user process generates the virtual address 11123456. Outline how the system establishes the corresponding physical location. 5(CO4)

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- (b) Under what circumstances do page faults occur. Describe the actions taken by the operating system when a page fault occurs. 5(CO4)
5. (a) Consider a system where free space is kept in a free-space list. Suppose that the Pointer to the free-space list is lost. Determine, can the system reconstruct the free space list. Explain your answer. 5(CO4)
- (b) Contrast the performance of the three techniques for allocating disk blocks (contiguous, linked and indexed) for both sequential and random file access. 5(CO4)
6. Summarize features and applications of RT Linux. 10(CO5)

